
URC
ULTIMATE ROBOTICS CHAMPIONSHIP

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Official Clarifications

Any clarification, amendment, or update to this rulebook shall be communicated only through the official URC communication channels.

Participants must rely only on the latest officially published version of the rulebook.

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1. GENERAL INFORMATION

1.1 Competition Overview

The Ultimate Robotics Championship 2026 (URC 2026) is a robotics competition that challenges participants to design, develop, and operate a multifunctional robotic system.
The competition evaluates key robotic capabilities, including:

- Autonomous navigation
- Line following
- Maze solving
- Obstacle navigation
- Object manipulation
- Precision and reliability

All participating robots must comply with the technical, operational, and safety requirements specified in this rulebook.

1.2 Competition Format

URC 2026 will be conducted in two stages.

Day 1 — Qualification Round

All eligible teams will participate in four challenges:

1. Line Follower
2. Maze Solver
3. Obstacle Course
4. Pick and Place

Each challenge carries a maximum of 50 points, making the maximum Day 1 score 200 points.
Teams will be ranked based on their cumulative Day 1 score. **Top 10 teams** will advance to the Final Round.

Day 2 — Final Round

Qualified teams will compete in advanced versions of the competition challenges. Day 2 will determine the final rankings and winners of URC 2026.

2. ELIGIBILITY AND TEAM REQUIREMENTS

2.1 Eligibility

Participation is open to students currently pursuing an undergraduate programme at a recognised educational institution.

2.2 Team Composition

Each team must consist of 2 to 5 registered members, including one Team Captain.
A participant may be registered with only one team.

2.3 Team Captain

Each team must designate one member as the Team Captain.
The Team Captain shall serve as the official point of communication between the team and the organizing committee.

2.4 Team Registration and Participation

All team members must be officially registered before participating in the competition.
Only registered team members are permitted to represent and operate the team's robot during the competition.

3. GENERAL ROBOT REQUIREMENTS

3.1 Robot Dimensions

The maximum permitted dimensions of the robot, including the robotic arm and attachments, are **200 mm × 150 mm × 120 mm** (L × W × H). An allowed dimensional error of +5 mm is specified.

3.2 Weight

The maximum permitted robot weight is **2500 g**.

3.3 Construction Requirements

Robots must be self-constructed or significantly modified by the participating team. Off-the-shelf robots are not permitted unless they have been heavily customised by the participants.

3.4 Permitted Modifications

The robot must comply with the specified size and weight requirements, and any modifications must remain within the limits permitted by the competition rules.

3.5 Autonomous and Manual Modes

Teams must ensure that their robots can switch between autonomous and manual modes of operation.

3.6 Robotic Arm Requirements

The robotic arm must:

- Be manually controllable using a remote control or interface provided by the team.
- Maximum payload provided: **150 g for Day-1.(40 x 40 x 40 mm Cuboid)**
- Maximum payload provided: **150 g for Day-2. (Any shape)**
- Have at least 3 degrees of freedom, including a jaw.
- Have shielded joints and moving parts.
- Include an emergency stop mechanism.

3.7 Power System and Battery

- The robot should be battery powered.
- Batteries must be securely mounted.
- Max Operating voltage should not exceed **16.8V DC(4S).**
- All wiring must be insulated and properly secured.
- Each robot must have an accessible and clearly marked power switch.
- Teams are permitted to bring spare batteries.

3.8 Mechanical Safety

Robots must:

- Not contain open flames or flammable materials.
- Be operated in a controlled manner.
- Avoid sudden or unpredictable movements.
- Be stopped immediately if they pose a danger or when instructed by an official.

4. COMPETITION PROCEDURES

4.1 Technical Inspection
All robots and systems will be inspected before the competition. Non-compliance with the rules may result in disqualification.

4.2 Start Procedure
The robot must start from the designated starting point.

4.3 Timing
The robot must complete the course in the least possible time.

4.4 Manual Intervention
Manual correction is permitted as specified in the respective challenge rules.

4.5 Restart Policy
A restart may be permitted only if the competition is interrupted due to an issue beyond the team's control, as determined by the organizing committee or officials.
A restart shall not be granted for failures or errors caused by the team's robot, including mechanical, electrical, or programming issues.

4.6 Repairs and Maintenance
The robot must be stopped immediately if it poses any danger or if instructed by an official. Any identified issues must be addressed promptly to maintain safety and functionality.

NOTE: The gap between two challenges will of 3 min.

5. DAY 1 CHALLENGES

5.1 Line Follower

Objective

The objective of this challenge is to assess the robot's ability to autonomously follow a black line (#000000) on a surface while demonstrating precise navigation and control.

Setup

The course consists of a clearly defined black line on the surface. The track may include:

- Straight sections
- Curves
- Sections with varying widths
- Sharp turns
- Intersections

The track is designed to challenge the robot's line-following capability without causing damage to the robot.

Requirements

The robot must be equipped with a line-following sensor.

Procedure

The robot must start from the designated starting point and autonomously follow the black line to the finish line.

The robot should navigate through curves and handle sudden changes in direction while remaining on the line.

The track includes intersections where the robot must continue following the black line without deviating from the designated path.

Performance

The robot must complete the course in the least possible time. Performance will also be assessed based on how accurately the robot follows the black line and handles the challenges of the course.

5.2 Maze Solver

Objective

The objective of this challenge is to evaluate the robot's ability to autonomously navigate through a maze or grid-like environment.

Setup

The track consists of a grid or mesh pattern with physical barriers. The maze may include elements such as:

- Junctions
- Dead ends
- Loops

The robot is required to navigate through the maze or grid to complete the task.

Requirements

The robot must be equipped with sensors capable of detecting the grid walls or barriers. Suitable sensing methods may include cameras, proximity sensors, or colour sensors.

Procedure

The robot must autonomously navigate through the grid or maze without human intervention.

The robot may follow the grid walls or use another suitable navigation method to move through the maze.

Performance

The robot's performance will be assessed based on its ability to successfully navigate and complete the maze efficiently.

5.3 Obstacle Course

Objective

The objective of this challenge is to evaluate the robot's performance while navigating through an obstacle course containing physical obstacles and challenges.

Setup

The obstacle course may include physical barriers such as:

- Ramps
- Cones
- Gates
- Narrow passages
- Tight turns
- Elevation changes

The course is designed to test the robot's manoeuvrability, control, and ability to navigate around physical obstacles.

Control Requirement

The robot must be manually controlled while following the designated marked trace through the obstacle course.

Obstacle Navigation

The robot must successfully navigate through all designated obstacles, including tight turns, ramps, and other physical challenges.

The robot should avoid collisions with or knocking over obstacles while navigating the course.

Performance

The robot must complete the obstacle course in the least possible time. Performance will be assessed based on successful navigation, collision avoidance, and the robot's ability to handle the obstacles.

5.4 Pick and Place

Objective

The objective of this challenge is to evaluate the robot's ability to pick up and arrange blocks according to a specified pattern.

Setup

The challenge takes place in a designated arena containing:

- A starting area for the robot
- A target area for arranging the blocks
- Six uniform blocks placed randomly within the arena

Pattern

A unique pattern and arrangement of the blocks will be assigned to each team at the start of the task.
The required pattern will be revealed at the beginning of the round.

Procedure

The robot must start from the designated starting area.
The robot must pick up and place the blocks one at a time.
The blocks must be arranged according to the pattern provided for the challenge.

Performance

The objective is to arrange the blocks accurately according to the required pattern.

6. DAY-1 SCORING

6.1 Line Follower Scoring

Scoring Component	Maximum Points
Successfull Completion	20
Checkpoints	20
Time Performance	0-10

Checkpoints

The course contains 4 checkpoints.

Each successfully cleared checkpoint shall be awarded 5 points.

Successful Completion

The robot shall be awarded 20 points for successfully reaching the finish line.

Time Performance

The challenge time limit is 5 minutes (300 seconds).

Time points shall be calculated as:

$$\text{Time Score} = 10 \times (300 - T) / 300$$

Where T is the robot's recorded completion time in seconds.

A robot that completes the challenge in less time shall receive a higher time score.

The maximum Time Score shall be 10 points.

eg. If you complete the challenge in 200 sec, then Time score will be:

10 x (300 - 200)/300 = 3.3 points

Manual Course Correction

If manual course correction is permitted, 3 points shall be deducted for each correction.

6.2 Maze Solver Scoring

Scoring Component	Maximum Points
Successfull Completion	20
Time performance	30

Successful Completion

The robot shall be awarded 20 points for successfully reaching the designated finish point.

Time Performance

The challenge time limit is 5 minutes (300 seconds).

Time points shall be calculated as:

$$\text{Time Score} = 30 \times (300 - T) / 300$$

Where T is the robot's recorded completion time in seconds.

The maximum Time Score shall be 30 points.

If the robot does not successfully complete the challenge, no completion or time points shall be awarded.

Manual Course Correction

If manual course correction is permitted, 3 points shall be deducted for each correction.

6.3 Obstacle Course Scoring

Scoring Component	Maximum Points
Successfull Completion	20
Checkpoints	20
Time Performance	0-10

Checkpoints
The course contains 4 checkpoints.
Each successfully cleared checkpoint shall be awarded 5 points.

Successful Completion
The robot shall be awarded 20 points for successfully reaching the finish point.

Time Performance
The challenge time limit is 5 minutes (300 seconds).
Time points shall be calculated as:

$$\text{Time Score} = 10 \times (300 - T) / 300$$

Where T is the robot's recorded completion time in seconds.
The maximum Time Score shall be 10 points.

Manual Course Correction
Manual driving is permitted and shall not result in a penalty.
If the robot requires physical repositioning or handling during the challenge, 3 points shall be deducted for each correction.

6.4 Pick and Place Scoring

Scoring Component	Maximum Points
Successfull Completion	20
Checkpoints	20
Time Performance	0-10

Checkpoints
The challenge contains 4 checkpoints.
Each successfully cleared checkpoint shall be awarded 5 points.

Successful Completion
The robot shall be awarded 20 points for successfully completing the required block arrangement.

Time Performance
The challenge time limit is 5 minutes (300 seconds).
Time points shall be calculated as:

$$\text{Time Score} = 10 \times (300 - T) / 300$$

Where T is the recorded completion time in seconds.
The maximum Time Score shall be 10 points.

Manual Course Correction
Manual driving is permitted and shall not result in a penalty.
If the robot requires physical repositioning or handling during the challenge, 3 points shall be deducted for each correction.

7. DAY-2 QUALIFICATION

Day 2 shall be the Final Round of the competition.
At the conclusion of Day 1, all teams shall be ranked based on their final Day 1 score.
The **top 10 teams** with the highest Day 1 scores shall qualify for Day 2.

Day 2 challenges are designed to be more demanding and require advanced robot capabilities, including:

- Speed
- Precision
- Autonomous decision-making
- Sensor integration
- Mechanical reliability
- Obstacle handling
- Navigation efficiency
- Object manipulation

8. DAY-2 FINAL CHALLENGES

8.1 Advanced Line Follower

- The robot will navigate a more complex line-following track containing challenging track elements and route decisions.
- The track may contain Sharp turns, Zig-zag sections, Dotted or interrupted lines, Complex junctions, Multiple paths, Rapid direction changes, Checkpoints
- The objective is to complete the course successfully while achieving the best possible time.

8.2 Advanced Maze

- The robot must solve a more complex maze than the Day 1 qualification maze.
- The maze may contain Multiple junctions, Dead ends, long routes, False paths, Complex intersections, Narrow passages, Route-selection challenges
- The robot must independently determine and follow an effective route to the FINISH.

8.3 Advanced Hurdle Race

- The final hurdle course will introduce more demanding obstacles and terrain conditions.
- The challenge may include, Inclined surfaces, uneven terrain, raised obstacles, Narrow passages, Direction changes, Multiple consecutive obstacles
- The robot must complete the course without external assistance.

8.4 Advanced Pick and Place

- The final pick-and-place challenge will require greater precision and control than the Day 1 challenge.
- The robot may be required to Navigate to the object location, Identify or locate the designated object, pick up the object securely, navigate to the target area, Place the object accurately, Complete the task within the specified time,
- The challenge will evaluate both the robot's navigation and manipulation capabilities.

9. DAY-2 SCORING

Each Day 2 challenge shall be scored using the same scoring structure, time calculation, penalties, and general scoring rules specified in **Section 6: Day 1 Scoring.**
Each Day 2 challenge carries a maximum of 50 points.

Challenge	Maximum Points
Advanced Line Follower	50
Advanced Maze	50
Advanced Obstacle Course	50
Advanced Pick and Place	50
Total	200

The final Day-2 score shall be calculated as:
Day 2 Score = Advanced Line Follower + Advanced Maze + Advanced Obstacle Course + Advanced Pick and Place
The maximum possible Day 2 score shall be **200 points.**

10. PENALTIES AND DISQUALIFICATION

Penalties may be applied for violations of competition rules, unauthorized intervention, unsafe operation, or failure to follow official instructions. Penalties shall apply to the relevant challenge, and the score shall not fall below 0 points.

Manual Course Correction

Where permitted, manual course correction requires referee approval. Each approved correction results in a 3-point deduction.

Unauthorized Intervention

Any unauthorized physical handling, repositioning, or adjustment of the robot during an active challenge may result in a penalty or termination of the attempt.

Technical Non-Compliance

A robot found to violate technical or safety requirements may be required to correct the issue before participation. Failure to comply may result in disqualification.

A team may be disqualified from an individual challenge or the entire competition for:

- Intentional damage to the arena or competition equipment.
- Intentional interference with another team or robot.
- Serious or repeated safety violations.
- Failure to comply with mandatory instructions from competition officials.
- Continued use of an unsafe or non-compliant robot.

Disqualification from an individual challenge results in 0 points for that challenge. Disqualification from the competition results in cancellation of all remaining attempts and removal from the final ranking.

11. TECHNICAL INSPECTION

All robots and associated systems must undergo technical inspection before participating in the competition.

The inspection shall verify compliance with the following requirements:

- Robot dimensions, including attachments and the robotic arm.
- Robot weight.
- Battery and power system safety.
- Secure mounting of batteries and components.
- Proper insulation and securing of electrical wiring.
- Accessible and clearly marked power switch.
- Mechanical safety of moving parts and joints.
- Shielding of robotic arm joints and moving components.
- Emergency stop mechanism for the robotic arm.
- Compliance with the specified autonomous and manual control requirements.
- Overall compliance with the technical and safety requirements of the competition.

A robot that does not meet the required specifications may be required to make corrections before being permitted to compete.

Failure to achieve compliance may result in disqualification from the relevant challenge or the competition.



12. TIE-BREAKING AND WINNER SELECTION

In the event that two or more teams obtain the same final score, the ranking shall be determined using the following criteria, applied sequentially.

12.1 Task Completion

The team that has successfully completed a greater number of competition challenges shall be ranked higher.

12.2 Penalty Points

If the teams remain tied, the team with fewer total penalty points shall be ranked higher.
All applicable penalties, including manual course corrections, shall be considered.

12.3 Autonomous Performance

If the teams remain tied, the team with the higher combined score in the autonomous challenges shall be ranked higher.
This includes:

- Line Follower
- Maze Solver
- Advanced Line Follower
- Advanced Maze

12.4 Cumulative Time Performance

If the teams remain tied, the team with the lower cumulative valid completion time across the applicable challenges shall be ranked higher.

12.5 Accuracy and Precision

If the tie persists, the team demonstrating better overall accuracy and precision shall be ranked higher.
This may include:

- Line adherence
- Navigation accuracy
- Obstacle handling
- Pick and Place accuracy

12.6 Technical Evaluation

If all preceding criteria remain equal, the judging panel shall evaluate the teams based on:

- Technical execution
- Robot reliability
- Engineering implementation
- Overall robot performance

The decision of the authorized judging panel shall be final.

13. Prize Distribution

Prize Distribution	
First Place	25000/-
First Runner Up	15000/-
Second Runner Up	10000/-

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